(a)
Semantic prior

Context C
Variables, data

LLM

Structured prior
Z_t

Compile

FiLM
ζ_t

Token bias
$\alpha_t \cdot l_{LLM}$

GP weights
$P_t(o)$

(b) Hybrid search

ζ_t | $\alpha_t \cdot l_{LLM}$ | $P_t(o)$

Policy search

GRU + FiLM
π_{θ_t}

Bias

$\mathbf{P}_t(a|s)$

s_0 [ ]
s_1 [+]
s_2 [+, x]
s_3 [+, x, sin(]
s_4
$x + sin(?)$

Evolutionary edits

s_2 Elite Queue
$x + sin(x)$
$x + log(x)sin(x)$
...

Crossover

Mutation

Candidates
$x + sin(x)$
...
$x + log(x)$

Evaluate
$R_{tr}(f)$
Fit constants · score

Elites
$x + sin(x)$
...
$x + cos(x))$

(c)
Experience feedback

Elite distillation
$x + sin\ x$
Policy update

Evidence audit
Lineages + validation
H_{t+1} a_{t+1}

Visual key
Prior Policy GP
Within round
Next round
$\ell_{LLM}(a) = d_t(a)$
s_i: expression prefix